

SkillSetu Sovereign National Platform

ACADEMICIAN & FACULTY LEADERSHIP: ARCHITECTURE, ALGORITHMS, UX & BMC SPECIFICATION

Document Identifier:	SKILLSETU-DOC-ACADEMICIAN-2026-V3	Classification:	Institutional Leadership Specification
Target Stakeholder:	Deans, HODs, Faculty Members, TPOs, Academic Councils	Policy Alignment:	NEP 2020 • NCrF Credit Harmonization • NAAC / NIRF
Backend Engine:	Node.js 20 LTS + TypeScript + Express + Prisma	Curriculum AI:	Google Gemini 2.5 Flash + Groq LPU Llama 3 70B

1. Executive Summary & Institutional Value Proposition

For universities, autonomous colleges, and faculty leaders, SkillSetu bridges the persistent divide between **academic coursework and dynamic corporate hiring demands**. Academic leaders gain an autonomous Curriculum Diff Engine that audits syllabi against live industrial job requisitions, an institutional attestation desk to endorse student competencies into the sovereign TrustLedger, faculty training programs for emerging tech upskilling, and enterprise research consultancies.

INSTITUTIONAL ACCREDITATION ENCLAVE

NEP 2020 & NCrF Institutional Compliance: Under National Education Policy 2020 mandates, institutions must provide multiple entry/exit pathways and vocational credit transfers. SkillSetu automates the mathematical credit translation formula ($\$Credits = \text{Notional Hours} / 30\$$), generating tamper-proof institutional transcripts for NAAC, NIRF, and NBA accreditation audits.

2. Technology Stack & Language Matrix for Academic Services

Layer / Tier	Technology / Framework	Technical Rationale & Architecture Role	Concrete Academician Benefit
Institutional UI	React 18.3.1 + TypeScript + Tailwind CSS	Clean administrative console optimized for faculty workflow, student roster reviews, and departmental audit reports.	Intuitive faculty portal requiring zero training; accessible on laptops, tablets, and classroom workstations.
Breadcrumbs Navigation	React Router DOM 6.26 + Breadcrumbs	Clear hierarchical navigation (SkillSetu > Academician > Consultancies / Students / Training).	Instant switching between student roster audits and corporate research consultancies.
Macro Analytics	Recharts 2.12.7 (SVG Compositor)	Macro-level competency distribution charts, cohort skill benchmarks, and placement readiness histograms.	One-click visual proof of student learning outcomes for institutional Board of Studies meetings.
Server Architecture	Node.js 20 LTS + Express 4.19 (TypeScript)	Stateless REST APIs enforcing allowedRoles: ['ACADEMICIAN'] role boundaries.	Secure, isolated departmental endpoints protecting student grading records.
Database Layer	Prisma 5.17 ORM + SQLite / Turso	Relational schema managing training_programs, consultancies, mentorship, and student rosters.	Transactional consistency across faculty endorsements and accredited training modules.
Curriculum AI	Google Gemini 2.5 Flash + Groq LPU Llama 3 70B	Dual-LLM semantic diff engine comparing university syllabi against corporate hiring requisitions.	Generates automated syllabus amendment proposals in seconds rather than months of manual committee reviews.

3. Mathematical Algorithms & Curriculum Governance Engines

Academic operations utilize advanced natural language processing and cryptographic verification algorithms:

Algorithm Name	Exact File Location	Mathematical / Technical Formulation	Academician Advantage
1. AI Curriculum Diff & Syllabus Harmonizer	services/aiService.ts (Lines 219-241)	Cross-correlates university course syllabus text against high-frequency enterprise recruiter requisitions: computes industryMatchPct, identifies missing critical skills, and flags obsolete legacy topics.	Empowers Board of Studies to modernize syllabi with empirical industry evidence, maximizing graduate employability.

2. National Credit Framework (NCrF) Credit Calculator	services/recommendation.service.ts & studentFeatures.controller.ts	$\text{Credits} = \frac{\text{Notional Learning Hours}}{30} \times \text{NSQF Level Factor}$ (Level 4.5 = 1.0, Level 5 = 1.1, Level 6 = 1.2, Level 7 = 1.3).	Automates academic credit calculations for internships, industry certifications, and lab assignments in full compliance with UGC guidelines.
3. Cryptographic Faculty Endorsement Oracle	controllers/studentFeatures.controller.ts (Lines 50–55)	Cryptographic signature combining Faculty ID, Institutional Public Key, Student ID, Competency Code, and UTC Timestamp into SHA-256 block hash.	Turns faculty members into trusted on-chain attestation oracles, cementing university credibility in the national skill registry.
4. Corporate Consultancy Synergy Matcher	routes/academician.routes.ts & CorporateExchange.tsx	High-dimensional cosine matching of faculty research publications and patents against industrial enterprise R&D; problem statements.	Connects university professors with high-value corporate consulting contracts and joint research grants.

4. End-to-End Academician User Experience (UX) Blueprint

The academician portal provides an efficient dashboard for student attestation, faculty upskilling, and industry consultancies:

Step	Interface / Route	Academician Action & Visual Feedback	Underlying System Telemetry
Step 1	Faculty Authentication /login	Faculty member logs in using institutional email; role guard verifies academic credentials and redirects to /academician.	JWT verified with ACADEMICIAN claim; loads departmental context.
Step 2	Academic Terminal /academician	Review departmental summary metrics: Enrolled Students, Total Skills Attested, Placement Parity Index, and Active Consultancies.	Aggregated queries over users, user_skills, and trust_ledger.
Step 3	Enrolled Student Roster /academician/students	Inspect departmental student cohorts, filter by branch/year, view individual skill portfolios, and endorse pending competencies.	Queries users where role='STUDENT' with joined skill tallies.
Step 4	Corporate Consultancies /academician/consultancies	Browse industry-funded R&D; projects, corporate technical challenges, and submit faculty consultancy proposals.	Loads active corporate exchange listings from consultancies table.
Step 5	Faculty Development /academician/training	Access accredited Faculty Development Programs (FDP), AICTE QIP workshops, and corporate sabbaticals in emerging tech.	Queries training_programs filtered by faculty level and topic.
Step 6	Student Mentorship /academician/mentorship	Review assigned student mentees, track capstone deliverables, and provide structured academic and career guidance.	Manages bilateral mentorship relationships in mentorship table.
Step 7	Institutional Analytics /academician/analytics	Export accreditation-ready PDF/Excel dossiers displaying cohort competency distribution, NCrF credit attainment, and NAAC metric criteria.	Recharts dynamically renders departmental attainment charts for Board of Studies.

5. API Key Architecture, Endpoints & Security Encapsulation

API Key / Parameter	Declared Location	Server Ingestion File	Operational Usage & Security Guard
GEMINI_API_KEY	backend/.env	services/aiService.ts (Lines 6-35, 219-241)	Executes the Curriculum Diff Engine comparing course syllabi against industrial requisitions. Completely hidden from browser client.
GROQ_API_KEY	backend/.env	services/aiService.ts (Lines 7, 42-69)	Backup inference engine ensuring instant syllabus gap analysis even during peak academic conference periods.
JWT_SECRET	backend/.env	middleware/auth.ts (Lines 8-18)	Guarantees institutional privilege separation; ensures student attestation oracles cannot be forged by unauthorized roles.

6. Societal & Educational Impact for Higher Education

SkillSetu revitalizes India's academic governance and accreditation standing:

- **NAAC, NIRF & NBA Score Enhancement:** Criterion 2 (Teaching-Learning & Evaluation) and Criterion 5 (Student Support & Progression) require verified outcome attainment metrics. SkillSetu provides auditable proof.
- **Eliminating Curriculum Stagnation:** Traditional syllabus revision cycles take 3–5 years. SkillSetu provides continuous, empirical industry gap telemetry, enabling agile semester-by-semester elective updates.
- **Faculty Industry Immersion:** Corporate consultancies and funded R&D; projects bridge ivory-tower academia with live enterprise challenges.
- **National Credit Harmony:** Seamless execution of the Academic Bank of Credits (ABC) under NEP 2020 through cryptographic NCrF credit attestation.

7. Business Model Canvas (BMC) — Academic Institution Dimension

BMC Building Block	Higher Education Institution Strategic Implementation
Customer Segments	Central & State Universities, IITs/NITs, AICTE-approved engineering institutions, autonomous private universities, medical/Ayush colleges.
Value Propositions	Automated NEP/NCrF credit compliance, empirical curriculum gap analysis, NAAC/NIRF accreditation dossier generator, faculty consulting marketplace.
Channels	Institutional SaaS portal, University ERP connectors, AICTE/UGC leadership conferences, Vice-Chancellor roundtables.
Customer Relationships	Institutional onboarding specialists, Dean/HOD governance dashboards, priority curriculum audit advisory.
Revenue Streams	Institutional Enterprise Licensing: Annual campus-wide subscription based on student enrollment; Accreditation Analytics Suite: Premium NAAC/NIRF reporting module; Corporate Consultancy Facilitation: 10% platform fee on funded industry projects.
Key Resources	AI curriculum diff engine, national skill ontology, cryptographic faculty endorsement oracle, accreditation report generator.
Key Activities	Curriculum taxonomy maintenance, university ERP integration, faculty training curriculum curation, accreditation compliance auditing.
Key Partnerships	AICTE, UGC, NAAC, NBA, State Higher Education Councils, NPTEL/SWAYAM, premier engineering institutes.
Cost Structure	Academic support team, curriculum AI inference compute, data security and DPDP compliance audits, university onboarding operations.